Evaluate each of the following integrals. For each integral, write down the integration technique you ended up using, and why you chose it. There are hints for a few of these on the next page. For every *indefinite* integral, check your answer by differentiating. This is part of the grading!

1. $\int \frac{\ln(t^2)}{t}\, dt$

☐Regrade

2. $\int xe^{2x}\, dx$

☐Regrade

3. $\int \cos^3 x \, \sin x \, dx$

☐Regrade

4. $\int_0^{\pi} x \sin(2x)\, dx$

□Regrade

5. $\int \cos^2 x \ \sin^3 x \, dx$

□Regrade

6. $\int_3^4 \frac{1}{2y+1}\, dy$

□Regrade

7. $\int_1^2 \frac{e^{\frac{1}{x}}}{x^2}\,dx$

□Regrade

8. $\int \frac{1}{\sqrt{1-x^2}}\,dx$

□Regrade

9. $\int \frac{1}{x^2+1}\,dx$

□Regrade

10. $\int_1^4 \frac{\cos(\sqrt{x})}{\sqrt{x}}\,dx$

□Regrade

11. $\int \tan^2 x\,dx$

□Regrade

12. $\int_0^1 e^{-2x}\,dx$

□Regrade

13. $\int (\ln x)^2 \, dx$

☐Regrade

14. $\int \frac{x}{x+1} \, dx$

☐Regrade

15. $\int \frac{x^2}{1+x} \, dx$

☐Regrade

16. $\int \tan^2 p \, \sec^2 p \, dp$

☐Regrade

17. $\int_2^{10} \frac{1}{\sqrt{5x-1}} \, dx$

☐Regrade

18. $\int_0^1 \arcsin t \, dt$

☐Regrade

19. $\int \frac{1}{\sqrt{x}(1+\sqrt{x})}\,dx$

☐Regrade

20. $\int \frac{1}{1+9r^2}\,dr$

☐Regrade

21. $\int \cos^2 x\,dx$

☐Regrade

22. $\int \frac{\sec^2 x}{\sqrt{\tan x}}\, dx$

□Regrade

23. $\int \cot x\, dx = \int \frac{\cos x}{\sin x}\, dx$

□Regrade

24. $\int \frac{2x}{\sqrt{1-x^4}}\, dx$

□Regrade

Hints

1. Algebra first!

□Regrade
5. $\sin^3 x = \sin^2 x \cdot \sin x$, and $\sin^2 x + \cos^2 x = 1$.

8. and 9. These are easy!

11. Look up a trig identity involving $\tan^2 x$ in the trig worksheets!

16. What is the derivative of $\tan x$?

20. Compare to Q9.

21. $\cos 2x = \cos^2 x - \sin^2 x = \cos^2 x - (1 - \cos^2 x) = 2\cos^2 x - 1$

24. Compare to Q8.